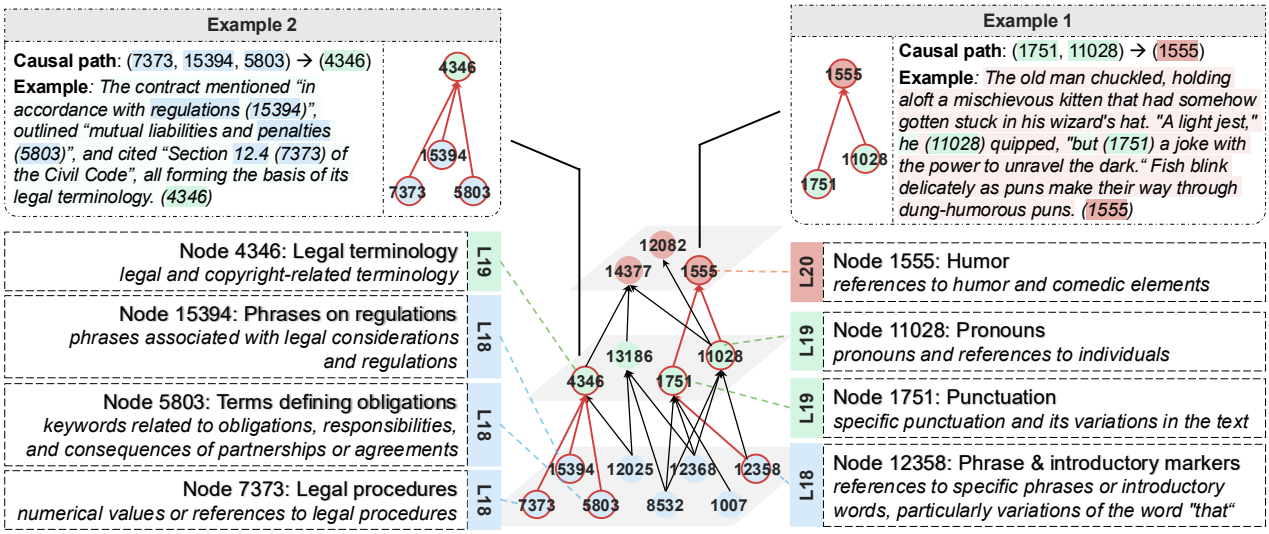

Figure H8. **The learned hierarchical concept graph for autoregressive language models.** We applied pretrained SAEs to Gemma-2-2b-it, partitioning tokens by sequence position (early, intermediate, final), then ran causal discovery across SAE features. The resulting graph reveals a hierarchical structure: high-level nodes (humorous tone, dog's role) caused by an intermediate node (pronouns and individual references). This shows the hierarchical discrete structure our theory posits does emerge in real transformer representations.

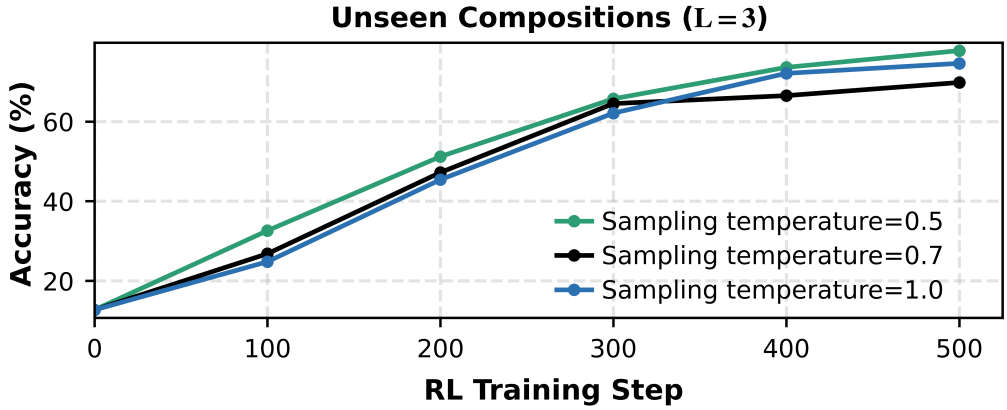

Figure H9. **Ablation study of the sampling temperature.** We follow the "$L = 2$, $C_{SFT} \cap C_{RL} = \emptyset$" setting in Figure 7 in the initial submission. Then we conduct an ablation study on sampling temperature (0.5, 0.7, 1.0). The results show that RL training is not extremely sensitive to this parameter.